

Automata Theory

Practice Problem Set 1

Topics: DFA, NFA, Regular Expressions, Automata Conversion,
DFA Minimisation and Regular Languages

Difficulty: Easy / Introductory

Number of Questions: 20

Section A: DFA Construction

Question 1

Construct a DFA over the alphabet

$$\Sigma = \{0, 1\}$$

that accepts all binary strings ending in 01.

For example:

- 01, 101, 1101 must be accepted.
- ϵ , 0, 10, 011 must be rejected.



Question 2

Construct a DFA over

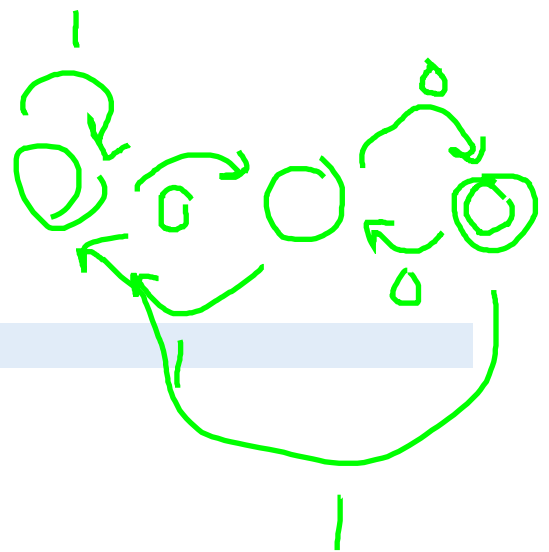
$$\Sigma = \{0, 1\}$$

that accepts all strings containing an even number of 0's.

The number of 1's may be arbitrary.

For example:

- ϵ , 1, 00, 1010 must be accepted.
- 0, 10, 000 must be rejected.



Question 3

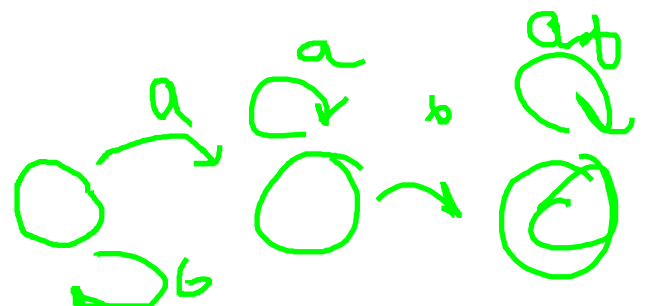
Construct a DFA over

$$\Sigma = \{a, b\}$$

that accepts all strings containing ab as a substring.

For example:

- $ab, aab, abba, babab$ must be accepted.
- ϵ, a, b, ba, bbb must be rejected.



Question 4

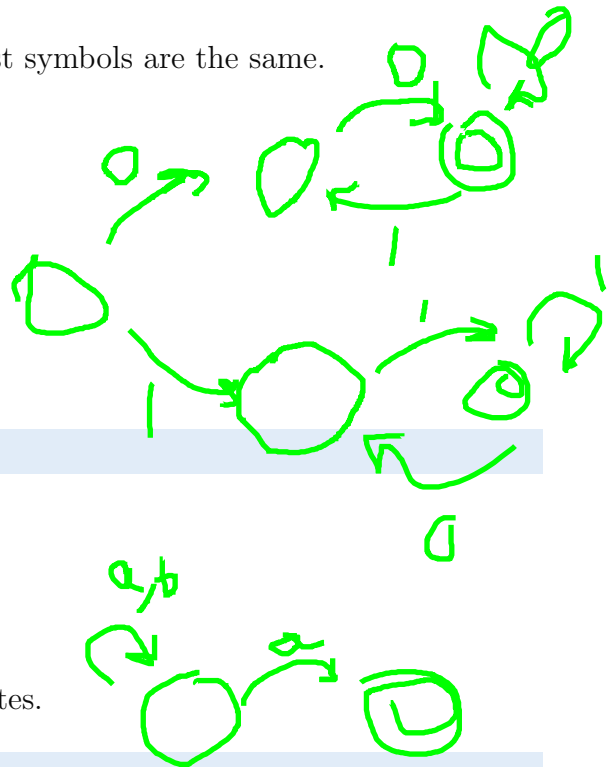
Construct a DFA over

$$\Sigma = \{0, 1\}$$

that accepts all nonempty strings whose first and last symbols are the same.

For example:

- 0, 1, 00, 11, 010, 1001 must be accepted.
- ϵ , 01, 10, 001, 110 must be rejected.

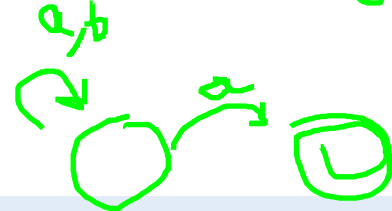
**Section B: NFA Construction****Question 5**

Construct an NFA over

$$\Sigma = \{a, b\}$$

that accepts all strings ending in a .

Try to construct an NFA with no more than two states.

**Question 6**

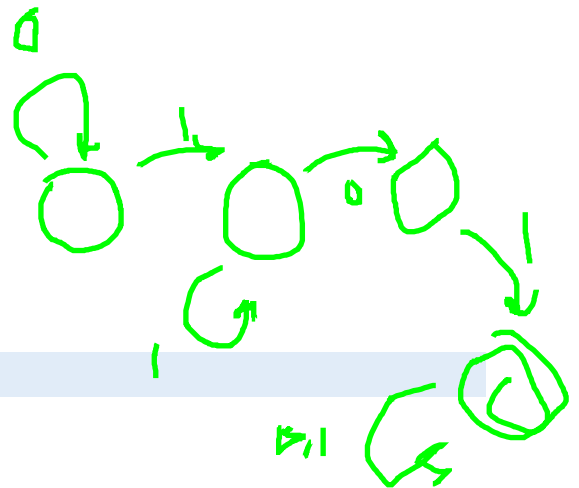
Construct an NFA over

$$\Sigma = \{0, 1\}$$

that accepts all strings containing 101 as a substring.

For example:

- 101, 0101, 1010, 11011 must be accepted.
- ϵ , 10, 100, 1111 must be rejected.

**Question 7**

Construct an NFA over

$$\Sigma = \{0, 1\}$$

that accepts all strings in which the third symbol from the right is 1.

For example:

- 100, 101, 110, 111, 0100 must be accepted.
- 000, 001, 10, 11 must be rejected.

Section C: Regular Expressions

Question 8

Write a regular expression over

$$\Sigma = \{0, 1\}$$

for the set of all binary strings ending in 00.

For example:

00, 100, 000, 1100

must belong to the language.

Question 9

Write a regular expression over

$$\Sigma = \{a, b\}$$

for the set of all strings containing exactly one occurrence of b .

For example:

- $b, ab, ba, aabaa$ must belong to the language.
- $\epsilon, a, bb, abba$ must not belong to the language.

Question 10

Write a regular expression over

$$\Sigma = \{a, b\}$$

for all strings that begin with a and end with b .

The shortest string in the language is ab .

For example:

- $ab, aab, abb, aabab$ must belong to the language.
- ϵ, a, b, ba, aba must not belong to the language.

Question 11

Write a regular expression over

$$\Sigma = \{0, 1\}$$

for all strings containing 01 as a substring.

For example:

- 01, 001, 010, 1011 must belong to the language.
- $\epsilon, 0, 1, 10, 111000$ must not belong to the language.

Question 12

For each regular expression below, describe its language in plain English and give two strings belonging to the language.

a)

$$(0 + 1)^*11$$

b)

$$a^*b^*$$

c)

$$(ab)^*$$

d)

$$0^*1(0 + 1)^*$$

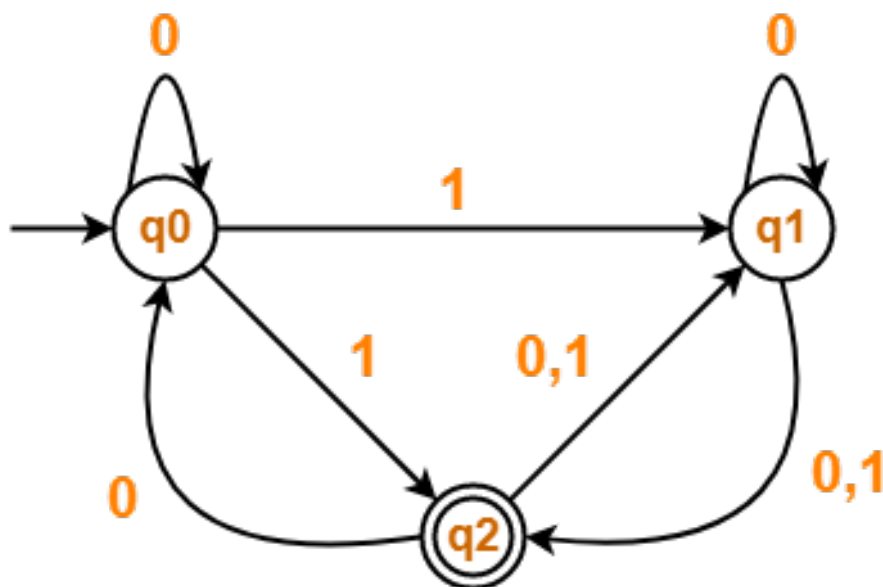
Also state whether ϵ belongs to the language of each expression.

Section D: DFA to NFA Conversion

Question 13

Consider the NFA shown below over the alphabet

$$\Sigma = \{0, 1\}.$$



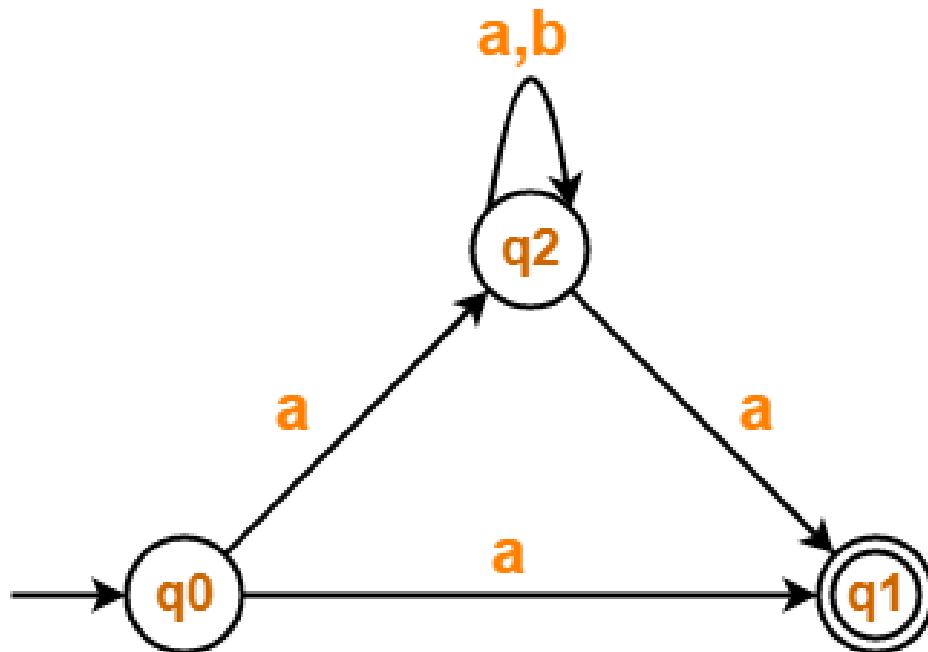
Convert the given NFA into an equivalent DFA

Question 14

Consider the NFA shown below over the alphabet

$$\Sigma = \{a, b\}.$$

- q_0 is the start state.
- q_1 is the only accepting state.

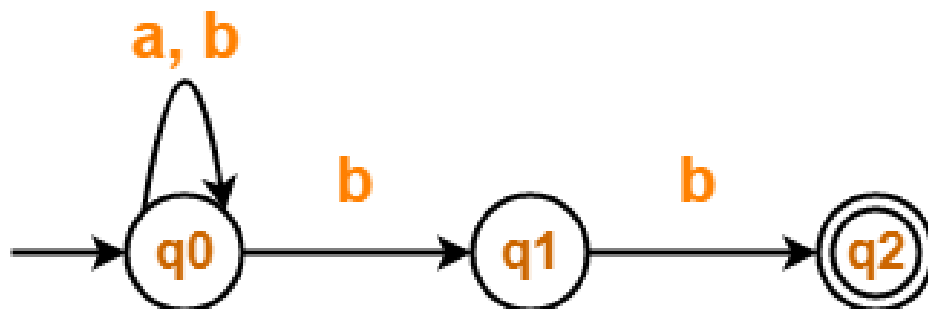


Convert this NFA into an equivalent DFA.

Question 15

Consider the NFA shown below over the alphabet

$$\Sigma = \{a, b\}.$$



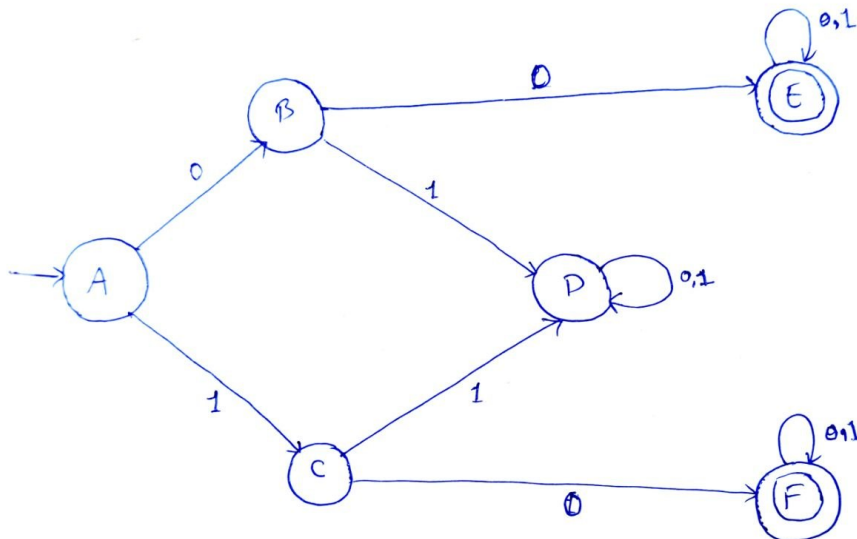
Convert the given NFA into an equivalent DFA using the subset construction method.

Section E: DFA Minimisation

Question 16

Minimise the following DFA.

- A is the start state.
- E and F are accepting states.



Perform the following:

1. Remove any unreachable states, if present.
2. Write the initial partition of the states.
3. Refine the partition until no further division is possible.
4. Identify all equivalent states.
5. Construct and draw the minimum-state DFA.
6. State the number of states before and after minimisation.

Question 17

Minimise the following DFA over

$$\Sigma = \{0, 1\}.$$

State A is the start state. States D and E are accepting.

State	Input 0	Input 1
<i>A</i>	<i>B</i>	<i>C</i>
<i>B</i>	<i>B</i>	<i>D</i>
<i>C</i>	<i>B</i>	<i>E</i>
<i>D</i>	<i>D</i>	<i>D</i>
<i>E</i>	<i>D</i>	<i>D</i>

Question 18

Minimise the following DFA.

State *S* is the start state. States *C* and *D* are accepting.

State	Input 0	Input 1
<i>S</i>	<i>A</i>	<i>B</i>
<i>A</i>	<i>A</i>	<i>C</i>
<i>B</i>	<i>A</i>	<i>D</i>
<i>C</i>	<i>C</i>	<i>C</i>
<i>D</i>	<i>C</i>	<i>C</i>
<i>E</i>	<i>E</i>	<i>E</i>

Perform the following:

1. Determine whether every state is reachable from *S*.
2. Remove all unreachable states.
3. Find all pairs of equivalent states.
4. Merge the equivalent states.
5. Write the transition table of the minimum DFA.
6. Draw the minimum DFA.

Section F: Regular Languages and Conceptual Understanding**Question 19**

For each language below, state whether it is regular. Give a brief reason.

1. The set of binary strings containing an even number of 1's.

2. The set of all strings over $\{a, b\}$ ending in ab .
3. The language
$$\{0^n 1^n \mid n \geq 0\}.$$
4. Every finite language.
5. The set of strings over $\{0, 1\}$ containing 010 as a substring.
6. The set of all strings over $\{a, b\}$ having length at most 5.

A complete pumping-lemma proof is not required. A short explanation based on finite memory, DFA construction, regular expressions or a known result is sufficient.

Question 20

Answer each statement as **True** or **False** and briefly justify your answer.

1. Every language accepted by an NFA can also be accepted by a DFA.
2. Every DFA can be regarded as an NFA.
3. An NFA may have more than one transition for the same state and input symbol.
4. A complete DFA must have exactly one transition from every state for every input symbol.
5. An NFA may have no transition for a particular state and input symbol.
6. Every finite language is regular.
7. The union of two regular languages is regular.
8. The intersection of two regular languages is regular.
9. The concatenation of two regular languages is regular.
10. The complement of a regular language is regular.
11. Every regular language can be represented using a regular expression.
12. Every regular expression represents a regular language.
13. An NFA is more expressive than a DFA.
14. An NFA may require fewer states than an equivalent DFA.
15. Two different regular expressions may represent the same language.

End of Practice Problem Set